IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-041

1. Topic (12 words max)

"The Guardian": A Modular AI for Proactive Human-Layer Cybersecurity on Social Media.

2. Research group the project belongs to

IAS - Information Assurance & Security

3. Specialization of the project belongs to

Cyber Security (CS)

4. If a continuation of a previous project:

Project ID	N/A
Year	N/A

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The contemporary digital landscape has shifted the cybersecurity paradigm from purely technical threats to human-centric vulnerabilities. Social media platforms, while fostering connection, have become primary vectors for a range of human-layer attacks that exploit psychology, trust, and cognitive biases (Ghafir et al., 2018). These threats include coordinated misinformation campaigns that erode public trust, hate speech that incites violence and harassment, sophisticated social engineering scams leading to financial loss, and the unintentional disclosure of Personally Identifiable Information (PII) that exposes users to further risks (Al-Rimy et al., 2018).

Current solutions are often inadequate. Platform-native moderation is reactive, typically acting only after harm has occurred and content has been reported. Existing security tools like antivirus software are not designed to interpret the nuances of human language and intent. Consequently, the average user is left unequipped to navigate this complex threat environment in real-time. They lack a proactive, non-intrusive mechanism that can guide their judgment without censoring their experience or violating their privacy.

This research addresses a critical gap: the need for an integrated, user-centric defense system that empowers individuals against a spectrum of linguistic threats. The core research problem is to investigate how state-of-the-art Natural Language Processing (NLP) models can be effectively trained and deployed in a modular architecture to provide real-time, context-aware analysis and gentle guidance to users, thereby enhancing their personal cybersecurity posture on social media platforms.

References:

- Al-Rimy, B. A. S., Maarof, M. A., & Shaid, S. Z. M. (2018). Ransomware threat success factors, taxonomy, and countermeasures: A survey and research directions. *Computers & Security, 74*, 144-166.
- Davidson, T., Warmsley, D., Macy, M., & Weber, I. (2017). Automated Hate Speech Detection and the Problem of Offensive Language. *Proceedings of the International AAAI Conference on Web and Social Media, 11*(1).
- Ghafir, I., Saleem, J., Hammoudeh, M., Faour, H., Prenosil, V., & Jaf, S. (2018). Security threats to critical infrastructure: the human factor. *The Journal of Supercomputing, 74*(10), 4986-5002.
- Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., & Choi, Y. (2019). HellaSwag: Can a Machine Really Finish Your Sentence? *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution, "The Guardian," is a modular AI-powered system implemented as a non-intrusive browser extension coupled with a powerful backend API. The system is designed to act as a friendly guide, analyzing text content on social media pages in real-time without blocking or censoring.

The browser extension frontend will be a lightweight client that scrapes text from posts, comments, and messages. This text is sent to the Guardian's backend API. The backend, built using Python and FastAPI, serves as the system's "brain" and hosts four distinct, specialist AI modules. Each module is a fine-tuned Transformer-based NLP model responsible for a specific threat vector: Hate Speech, Misinformation, Phishing/Scams, and PII/OpSec Risks.

Upon analysis, the API returns a structured JSON response to the extension. The extension then overlays a subtle, color-coded icon next to the analyzed content. Hovering over the icon reveals a concise, informative tooltip explaining the potential risk (e.g., "This post contains language consistent with hate speech," or "This message creates false urgency, a common scam tactic."). This architecture ensures user privacy and freedom while providing timely, actionable intelligence to fortify their human-layer defenses.

Conceptual Diagram:

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+-------------------------+     (1) Scrapes Text     +----------------------------+
| User on Social Media    |------------------------>|  GUARDIAN BROWSER EXTENSION  |
| (e.g., Facebook, Twitter)|                        | (Lightweight Frontend Client) |
+-------------------------+     (4) Displays         +----------------------------+
           ^                  Non-intrusive Icon/Alert           |
           |                                                     | (2) Sends Text via HTTP
           |                                                     v
           |                                         +----------------------------+
           |                                         |     GUARDIAN BACKEND API     |
           |                                         |        (The "Brain")         |
           |                                         |----------------------------|
           |                                         | Module 1: Hate Speech AI     |
           | (3) Receives JSON Analysis              | Module 2: Misinformation AI  |
           +-----------------------------------------| Module 3: Scam/Phishing AI   |
                                                     | Module 4: PII/OpSec AI       |
                                                     +----------------------------+
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7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Domain Expertise: The project is rooted in Cyber Security, specifically focusing on Information Assurance, Social Engineering, and Human-Computer Interaction (HCI) security principles.

Knowledge Requirements: The team will need to acquire specialized knowledge in several key areas.

- **Core Programming & Data Science:** Proficiency in Python, including libraries essential for data manipulation and analysis such as Pandas and NumPy.
- **Artificial Intelligence / Machine Learning:** A deep understanding of Natural Language Processing (NLP) is paramount. This includes traditional techniques (TF-IDF) for baseline models and advanced concepts like Transformer architectures (e.g., BERT, DistilBERT), transfer learning, and model fine-tuning.
- **AI Development Tools:** Hands-on expertise with the Hugging Face ecosystem (Transformers, Datasets, Tokenizers) and a deep learning framework like PyTorch or TensorFlow.
- **Software Engineering:** Skills in backend API development using a framework like FastAPI or Flask, and frontend development for creating the browser extension using JavaScript, HTML, and CSS.
- **Version Control:** Competency in using Git and GitHub for collaborative development and code management.

Data Requirements: The project's success relies on high-quality, labeled datasets for training and validating each AI module. We will primarily leverage publicly available, academic datasets from sources like Kaggle, Papers with Code, and university data repositories. Examples include the Davidson et al. hate speech dataset and the LIAR dataset for misinformation. For more specific tasks like social media scam detection or PII leakage, we may need to perform data augmentation or create a smaller, custom-labeled dataset based on established examples and patterns.

8. Objectives and Novelty

Main Objective To design, implement, and evaluate "The Guardian," a modular, AI-powered system that proactively assists users in identifying and understanding human-layer cybersecurity threats on social media in a non-intrusive manner.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Herath, H. M. G. D. K. (IT22361486)	To develop a module for detecting unintentional PII disclosure and Operational Security (OpSec) risks in user-generated text.	• Define module interface & API specs; lead integration, • Research & implement a Named Entity Recognition (NER) model to find PII, • Train a classifier for OpSec risk patterns.	The proactive application of NER to provide pre-emptive OpSec advice *before* a user posts sensitive information, moving beyond simple data loss prevention to user education.
Fernando, M. R. I. (IT22366054)	To develop and evaluate a specialized AI module for the detection of hate speech and toxic language, including	• Research hate speech linguistics, • Collect & clean relevant datasets • Fine-tune a Transformer model for toxicity classification; • Analyze model biases and fairness.	Moving beyond basic keyword flagging to detect implicit and context-dependent toxicity (e.g., "dog whistles"), improving accuracy while reducing false positives on legitimate critiques.

	subtle and coded forms.		
De Alwis, H. P. G. H. S. (IT22317926)	To create and validate an AI module capable of identifying linguistic patterns indicative of misinformation and psychological manipulation.	• Research propaganda techniques & logical fallacies; • Gather datasets of verified misinformation; • Train a model to classify text based on manipulative markers (e.g., loaded language, false urgency).	A focus on identifying the *persuasive techniques* of disinformation rather than just assessing factual accuracy, providing a more robust defense against novel "fake news" narratives.
Vishwa J R (IT22916112)	To build and test an AI module for the detection of text-based social engineering, phishing, and scam attempts prevalent on social media platforms.	• Research the linguistics of online scams; • Curate a dataset of scam messages; • Train a classification model to identify patterns like grooming, impersonation, and illicit calls-to-action.	The model is specifically tailored to the short-form, informal, and often image-adjacent text of social media, a domain distinct from traditional, long-form email phishing filters.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Herath, H. M. G. D. K. (IT22361486)	This component directly addresses the Cyber Security principle of **Confidentiality**. By developing a module to detect PII and OpSec risks, the research focuses on preventing data leakage and strengthening a user's personal security posture against social engineering attacks that leverage publicly available information.
Fernando, M. R. I. (IT22366054)	This research complies with Cyber Security by focusing on **Threat Analysis and Mitigation**. Hate speech and online harassment are significant threat vectors. Developing an automated system to identify this threat contributes to creating safer online environments and reducing the psychological impact of cyberbullying.
De Alwis, H. P. G. H. S. (IT22317926)	This module is centered on the Cyber Security specialization by tackling the problem of **Information Integrity**. Misinformation campaigns are a form of information warfare. This research explores automated defenses to help users verify the integrity of the information they consume, thus mitigating the risk of manipulation.
Vishwa J R (IT22916112)	This component is a classic application of Cyber Security, focusing on **Fraud Prevention and Anti-Phishing**. It involves developing a specialized defense mechanism against fraudulent schemes. The research contributes directly to protecting users from financial loss and identity theft initiated via social media platforms.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Kavinga	Abeywardena	
Co-Supervisor	Mr.	Deemantha	Siriwardana	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*Important:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.